



Uncertainty and purpose of measurements of carbon sequestration in soil[☆]

Aja Watkins^{ID}

University of Wisconsin-Madison, 600 N Park, St, Madison, WI, 53706, USA

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ABSTRACT

Many philosophers of measurement agree that measurements and measurement uncertainty need to be evaluated by their adequacy for a particular purpose. This paper uses an analysis of measurements of carbon in soil, which are used to support carbon trading and carbon offsetting, to test the adequacy-for-purpose method of measurement evaluation. In particular, evaluating the soil carbon case requires not just matching measurement practices to aims, but also being able to evaluate the aims of carbon trading or offsetting and being able to tell whether the measurement practices are (in)adequate for those more-ultimate aims.

1. Introduction

Philosophers of measurement are increasingly advocating for a view according to which measurement processes and measurement outcomes should be evaluated relative to a purpose or set of purposes, rather than evaluated in some context-independent sense. For example, the importance of features of measurement outcomes such as their precision may vary by context and there is nothing wrong with that; very precise measurement outcomes are not required for all purposes.

The acceptable degree of uncertainty or the required degree of accuracy of a measurement procedure or instrument are also subject to context-relative assessment. In fact, some scholars have gone so far as to argue that the very meanings of accuracy and truth are relative to a contextualized purpose [1,2]. In general, following Chang's [3] statement of the "problem of nomic measurement," philosophers have rejected "foundationalism" about measurement according to which we should assess measurement results by their correspondence with "real" quantities (as a traditional notion of accuracy would suggest), because we do not have access to "real" quantities independent of the measurement procedures that we would be trying to evaluate in the first place. Instead, many of us now prefer a pragmatist or coherentist view of measurement evaluation that says we can only evaluate measurement instruments, procedures, and outcomes within a context that also contains other scientific practices and aims.

After rehearsing the arguments for an "adequacy-for-purpose" view of measurement evaluation in Section 2, I will put these insights to the test, using a case study involving the measurement of carbon in soils (Section 3). Carbon is measured in soils for a variety of purposes, but I will focus on how these measurements are used to support carbon trading and offsetting. In brief, some carbon trading and offsetting

schemes involve incentives for land owners to treat their soils in particular ways, ways that are thought to encourage the soils to sequester carbon. In order to know how much carbon is being sequestered by which soil treatments, we have to be able to measure carbon in the soil both in the absence and presence of those treatments. However, these measurements are highly uncertain.

An analysis of this case study contributes to the argument for a series of upshots (Section 4). First, contrary to intuitive expectations about the relationship between uncertainty and aim-facilitation, the uncertainty of these soil carbon measurements does not necessarily impede the achievement of their aim, depending on the aim. According to the adequacy-for-purpose view of measurement evaluation, then, nothing has gone wrong: the measurements are adequate for their purposes, in spite of their uncertainties. Second, though, I will argue that we might still want to criticize the combination of scientific practices that involves using soil carbon measurements for these aims; doing so will require going beyond an adequacy-for-purpose view. Third, then, I identify some room for improvement in the adequacy-for-purpose view: namely, we need to not only assess our measurement practices in light of our purposes and our purposes in light of our measurement practices, but we also need to assess our purposes in other ways. In the soil carbon case, different ultimate purposes for the carbon trading and offsetting schemes (from mitigating global climate change to exonerating corporations from culpability) yield different verdicts on whether the carbon trading and offsetting purposes are okay to pursue.

Ultimately, then, this paper contributes to existing literature in philosophy of measurement and continues a tradition of philosophical engagement in measurement theory facilitated by the Society for the Study of Measurement (SSM) conferences (e.g., [4–22]), including contributions to *Measurement* (e.g., [23–27]).

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E-mail address: aja.watkins@wisc.edu.

2. Measurement evaluation

Many philosophers of science endorse an “adequacy-for-purpose” view, according to which scientific practices and tools, including those related to measurement, should be assessed relative to their purposes.

For example, in philosophy of modeling, Parker [28–30] has influentially argued that models should be evaluated not by how accurately they represent their target systems but instead by how adequate they are for specified research purposes (see also [31,32]; cf. [33,34]). Her adequacy-for-purpose view of model evaluation helps make sense of the pervasive presence of idealizations and abstractions in models [35–39]. If one thought that the better model was always the one that was more closely analogous to its target system, then the usefulness of idealizations and abstractions would be mysterious. According to the adequacy-for-purpose view of model evaluation, by contrast, idealizations and abstractions will be beneficial insofar as they help models serve certain purposes.

The adequacy-for-purpose view can easily be applied to scientific practices and tools other than modeling, too. Importantly, Bokulich and Parker [40] have proposed a view of data evaluation according to which data, like models, should not be evaluated by how well they “mirror” reality, but instead by how well they serve the aims of science in its context.¹ They have, likewise, argued that data uncertainty should be evaluated by its adequacy for purpose [43] (see also [44]). Beyond the work on models and data, the adequacy-for-purpose criterion has now been connected to a variety of data- and model-related scientific practices, including citizen science [45,46], reconstructions of the deep past [47,48], ecological theories [49], the use of expert judgment [50] and more, and has been expanded from a narrow focus on the epistemic aims of science to a wider set of non-epistemic purposes [51,52].

If one is on board with an adequacy-for-purpose account of model and data evaluation, it is not difficult to see how a similar view of measurement evaluation could work. In part, this is because models, data, and measurement are interrelated scientific practices: obviously measurements are one way to generate data, but also measurements may be model-based [12,13,53,54] and data and models are symbiotically related [55–57].

The key insight from an adequacy-for-purpose view of measurement is that various features of measurement instruments, procedures, or outcomes should be evaluated relative to purposes and contexts, rather than in an absolute, de-contextualized way. Many important advances in philosophy of measurement have come from applying or extending the adequacy-for-purpose view. Philosophers of measurement thinking along these lines often take a particular feature of measurement that is presumed to be *always* good or beneficial – e.g., accuracy, precision, standardization – and use a case study to show that, given the aims and other aspects of the context in that case, those features do not straightforwardly make the measurement process better. For example, Jackson [21,24] has argued that precision and standardization of some measurements in medical sciences (e.g., measurements of volume of anesthesia, measurements of cervical dilation) better serve some purposes, namely, holding practitioners accountable when they make mistakes, than others, such as patient experience (see also [58]). Thompson [59] has also argued that, contrary to the assumption that measuring on an interval scale is always better than measuring on an ordinal scale, ordinal measurements of pain by self-report are perfectly adequate for at least some purposes (see also [60]). Additional work along similar lines challenges the necessary pursuit-worthiness of coherence between different measurements of the same

¹ There is interesting disagreement between Bokulich and Parker’s pragmatic-representational account of data and Leonelli’s relational view of data [41,42]. However, both views are opposed to a view of data on which data “mirror” reality, and both would endorse a broadly pragmatic method for data evaluation.

measurand [61,62], the value-freedom of measurement [23,63,64], the context-independence of validation [65,66], and the benefits of reusability of measurement results [67].

Despite its fruitfulness for the philosophy of measurement, the adequacy-for-purpose view is not beyond criticism or revision. Some recent work questions the scope and applicability of the adequacy-for-purpose view and points out some gaps in its analysis. Bocchi and Santana [68] argue that, despite some of the merits of an adequacy-for-purpose view of measurement evaluation, the view lacks any resources to evaluate the *processes* that produce measurements; for example, the democratization of some measurements is an important feature that should be evaluated as such but the complexity of these evaluations is not reflected in the relatively simple purpose-practice matching protocol suggested by an adequacy-for-purpose view. Additionally, Wallis [69] argues that an adequacy-for-purpose view of data evaluation fails to capture the division of labor that often occurs between those who produce data and those who use data. The producers are often the ones who make decisions that shape the data and, consequently, shape its potential adequacy for various purposes, but the users are often the ones setting the purposes. Wallis thereby diagnoses a possible reason why some data would fail to be adequate for their purposes, but also suggests that the decisions made in the process of producing data might be more complicated than the versions of an adequacy-for-purpose view of data evaluation currently on offer can handle.

What these and other criticisms of the adequacy-for-purpose view have in common is, first, a commitment to the pragmatist spirit of the adequacy-for-purpose view, but, second, a complaint that the adequacy-for-purpose view is not pragmatist *enough*. For example, Bocchi and Santana’s [68] criticism is inspired by other commitments of the philosophical pragmatist, such as a commitment to science as a social, contextual process; Wallis’ [69] criticism is also compatible with general insights from pragmatism, such as the sociality of science and the iterative nature of its development. Indeed, many philosophers of measurement have articulated a more profoundly and explicitly pragmatist view of measurement than the adequacy-for-purpose view has the resources to provide [1,3,42,44,70].² An important question, then, is whether an adequacy-for-purpose view of measurement really fulfills its pragmatist ambitions, and the view can be evaluated in that way.

This paper contributes to these efforts to assess how pragmatist the adequacy-for-purpose view of measurement really is. In particular, I will discuss how the current uncertainties associated with measurements of carbon in soil do not necessarily make these measurements less adequate for their purposes, contrary to a commonsensical view according to which uncertainty always reduces adequacy. An explanation of this measurement context, including a connection between uncertainty and adequacy, is the subject of the next section. Then, in Section 4, I will propose a method for evaluating these measurement procedures that goes beyond the simple adequacy-for-purpose view. A truly pragmatist method of measurement evaluation will require not only assessing fit between purposes and measurement procedures, but also must enable us to evaluate the purposes themselves.

3. Measuring carbon in soils

Increased concentrations of carbon dioxide and other greenhouse gases in the atmosphere are causing Earth’s climate to change to an extent that has already had and will continue to have profound effects on societies and ecosystems worldwide.

One way to (partially) address climate change is through carbon sequestration.³ Carbon sequestration involves removing carbon from

² What, exactly, a comprehensively “pragmatist” philosophy of science would look like is a complicated and controversial topic that is outside of the scope of this paper (although see [71–73] for helpful discussion).

the atmosphere, in order to ideally slow down or reverse the effects of its presence. There are many ways to sequester carbon, including some engineering-based solutions [81,82] and so-called “biosequestration” solutions [83].⁴ For example, photosynthesis removes carbon dioxide from the atmosphere and stores it in the biosphere, and is thereby one of the main drivers of the global carbon cycle on relatively short time scales [85]. However, carbon sequestration has not yet succeeded at preventing or reversing global climate change, in part because it is costly in the short term and does not have immediate benefits (monetary or otherwise). Carbon trading, carbon offsetting, or carbon credit programs have all been initiated in various ways to attempt to incentivize investment in carbon sequestration [86].

The kind of carbon sequestration program most relevant to this paper involves regenerative agriculture practices and their involvement in carbon offsetting schemes; indeed, Khangura et al. [87] list “Capturing soil carbon” as one of the main goals of regenerative agriculture (see also [88–91]). There are variations in how these programs work and exactly which agricultural practices are incentivized through them, but here is an example from the work of science journalist Popkin [92]: Companies which are looking to offset their carbon emissions (in this case, he gives examples of IBM, JPMorgan Chase, and Shopify) can pay a third-party company (in this case, a company called Indigo), which in turn pays for carbon to be sequestered by various means. Popkin spoke to farmers who were paid to use “no-till” regenerative agricultural practices in their fields – literally, paid to not till their fields rather than to till them – under the assumption that doing so would contribute to carbon sequestration. The idea behind no-till practices as a means to carbon sequestration is that by disturbing the soil less, the various chemical reactions that cause soil to emit carbon dioxide (e.g., via decomposition) are slowed down because less of the soil is exposed to oxygen. The farmers involved in these programs are paid by companies like Indigo commensurate with the amount of carbon that is calculated to be sequestered by practices like no-till.

If everything goes according to plan, carbon offsetting programs such as these are win-win-win-win: the corporation interested in offsetting gets to claim investment in environmental causes (presumably they have calculated that there is demand for this among their customers), the third-party company gets to take a cut of the funds, and the farmer gets paid to sequester carbon – plus, carbon sequestration benefits all of us via its role in climate change mitigation. However, there is some skepticism about using soil credits for these purposes [92,93]. According to Popkin, much of this skepticism boils down to an assessment that the measurements of carbon being sequestered by soils are too uncertain: “researchers say the science of how soils store and release carbon is too uncertain to support an industry claiming to be cooling the planet” [92] and “the biggest impediment to the widespread adoption of climate-friendly farming is the lack of a practical way to quantify the soil carbon gained through a regenerative tactic” [92] (see also [94,95]). These uncertainties have led many to admit that the potential benefits of carbon sequestration in soil are limited [96].⁵

³ There is some debate about whether carbon sequestration is a good idea, e.g., see [74,75]. To some extent these debates parallel debates about whether thinking about one’s carbon “footprint” makes any sense. For example, carbon footprinting has been criticized for assuming that carbon emissions really can be compensated for and disagreements remain about the political utility and best operationalization of the carbon footprint concept [76–79]. These criticisms of the carbon footprint concept have led to calls for refinements, such as using an “ecological footprint” concept instead [80].

⁴ Regarding “green” versus “gray” solutions to environmental problems in general, see [84].

⁵ One notable controversy has involved whether certain goals for soil carbon storage – in particular, a goal to increase soil organic carbon by 0.4% annually – is biogeochemically feasible or unrealistic; see [97–101] for discussion.

In order to evaluate the measurements of soil carbon, then, we need to account for their uncertainties [43,44,102]. What exactly makes these measurements so uncertain? I will now review the various sources of uncertainty involved, after which I will relate those sources of uncertainty to the purposes of the measurements.⁶

3.1. Sources of uncertainty

1. Sampling depth

Even before we try to start using soil carbon measurements to estimate how much carbon was sequestered by a given intervention, we have to be able to take a snapshot measurement of how much carbon is currently in one volume of soil. These snapshot measurements become part of the larger measurement process involved in measuring sequestered carbon by, for example, being used to compare carbon concentrations before and after an intervention was performed or by comparing carbon in soil treated one way with carbon in soil treated another way.

However, one major source of uncertainty with these snapshot measurements has to do with the depth at or over which the soil carbon is measured.

First, choosing *any* depth to use for sampling is going to have the problem that measurements of samples taken at that depth will not be able to distinguish between, on the one hand, carbon exchange between the soil and the atmosphere, and, on the other hand, carbon exchange between the soil within the sampling depth and the soil or other materials below that sampling depth. For example, one of the effects that *tilling* soil has is pushing crop residues deeper into the soil than they would have ended up in the absence of tilling [92]. “Most experts now believe that the practice redistributes carbon within the soil rather than increases it, though it can improve other factors such as water quality and soil health” [93]. This means that some amount of the carbon that ends up in surface soils as the result of no-till treatments might have been in the soil anyway (just at a deeper level than is measured) in the absence of these treatments (i.e., in a regular tillage condition). Calculating, then, the amount of carbon that is sequestered from the atmosphere as the result of no-till will have to involve subtracting the amount of carbon that is newly in the top layers of soil instead of deeper in the soil (rather than instead of in the atmosphere). But figuring out this quantity is difficult, contributing to the uncertainty of the total sequestration estimate.

Second, measuring carbon at a constant soil *depth* has to contend with other changes to the soil due to changes in treatment, especially changes in soil *density* [103]. In particular, no-till treatments increase the density of the topsoil by some (itself uncertain) amount compared to till treatments. Even if no carbon is sequestered from the atmosphere due to the conversion to no-till, the sampled soil after the conversion would be expected to contain more carbon than the sample before the conversion, just due to the increase in density (there is just a greater mass of soil in the sample after the conversion than before). So, the change in density has to be accounted for in the calculation of sequestered carbon; but, if we are uncertain about the extent to which the density has changed, it will be difficult to account for.

Choices related to sampling depth, then, come with multiple complications relating to how to deal with the possibility of carbon transfer below the sampling depth and having to account for other changes to the soil such as changes to its density. Issues related to depth

⁶ In what follows, I will not present *quantified* uncertainty estimates. Although these would be helpful, they often do not exist, or are themselves too uncertain to be of much use here. Soil scientists and others are working on quantifying these uncertainties – see Section 3.2 – but more work remains to be done. Thanks to an anonymous referee for encouraging me to spell this out in more detail.

and timescale of soil sampling have led to calls for longer-term and deeper-sampled studies [87,104].

Notice, in particular, that sampling depth as a source of uncertainty about how much carbon is in the soil at any given snapshot in time is a particularly *relevant* source of uncertainty only given a broader measurement context in which these measurements will be used to evaluate the relative success of carbon sequestration techniques. It is foreknowledge of how these samples will be used that causes concern about this source of uncertainty.

2. Counting counterfactual carbon

In order to support carbon trading and offsetting programs, the measurement we need is an amount of carbon that is in the soil *due to* a change in management practice. In order to know that, we need to know not only how much carbon is in the soil after the change in management practice (e.g., after conversion to a no-till regime) but also how much carbon *would have been* in the soil but for that change.⁷ This latter quantity is a counterfactual amount of carbon. Only by comparing the actual to the counterfactual amount of carbon can we know how much carbon has been sequestered due to the carbon trading or offsetting scheme; it is then that amount of increased carbon sequestered that can be taken “credit” for.

There are really several separate issues with comparing actual to counterfactual carbon. The first issue is that it is hard to measure nonexistent quantities. What would be required is some sort of carefully controlled comparison: two fields that are identical in all relevant respects except for one change in management, between which you could compare the amounts of carbon in the soil after some amount of time. This setup is not impossible in principle, but in practice it is very challenging to set up or identify fields which count as “identical in all relevant respects except for one”, in part because it is difficult to know which respects are the relevant ones (e.g., see [87,105–107]) and in part because real plots of land are messy and it is difficult to control their properties experimentally.

The second issue related to measuring counterfactual soil carbon is that it is unclear how often farmers are changing their management practices in response to getting paid to sequester carbon and how often they are, instead, just utilizing programs that incentivize management practices that they are already implementing or planning to implement for independent reasons [92]. Of course, the farmers can hardly be blamed for doing this; if I learned of ways I could make more money doing my job exactly the way I was already planning on doing it, I would take advantage of that, too! The problem is that the completely rational actions of farmers make it difficult to say how much, if any, carbon is being sequestered by these programs. If the farmers were going to use no-till methods anyway, for example, then no *additional* carbon is going to be sequestered if they are paid extra to use no-till methods. Carbon crediting is predicated on the assumption that additional carbon will be sequestered through these programs, and, if it is not, then they are not working.

Third, although there is some evidence to suggest that the amount of carbon in soil after treatments like no-till does increase, it is very hard to distinguish between increases in soil carbon due to carbon sequestration and increases in soil carbon due to carbon retention – carbon that, but for the change in management, would have been lost to the environment sooner [108]. Depending on exactly how one thinks about carbon budgeting, sequestering carbon and retaining carbon that otherwise would have been emitted either are or are not equivalent.

⁷ It might be helpful to think of an analogy here related to burning calories: when you want to know, for example, how many calories you burned due to a particular exercise program, part of what has to be calculated is how many calories you would have burned in the absence of that exercise. Oftentimes, this is difficult to know for sure. Thanks to Roberta Millstein for pointing out this analogy.

Finally, ideally we would want to compare strategies such as no-till to other alternative means of carbon sequestration (even other regenerative agricultural techniques such as the use of cover crops) [109,110]. And we would want to evaluate any trade-offs involved in pursuing one of these strategies over another; for example, some evidence suggests that although no-till treatments do sequester carbon, they increase emissions of nitrous oxide (N₂O), another powerful greenhouse gas [111,112]. No-till agriculture may also increase the use of herbicides, the environmental effects of which would have to be evaluated against purported carbon sequestration benefits [113,114].⁸ In general, in the absence of a principled selection of counterfactual scenarios to use for comparing these different strategies, it is difficult to evaluate their relative effectiveness at climate change mitigation.

In summary, there is uncertainty about how much carbon would have been stored in the soil in the absence of incentives to sequester carbon using methods like no-till agriculture. Only by comparing actual to counterfactual carbon storage can we know how much carbon is being sequestered due to these incentives. Again, importantly, uncertainty about counterfactual carbon becomes a *relevant* source of uncertainty when situated within a larger measurement context; we do not just want to know the amount of carbon sequestered compared to other scenarios for its own sake, but, rather, to be able to assess the efficacy of various carbon trading programs at actually sequestering carbon.

3. Longevity of carbon storage

A third major source of uncertainty related to soil carbon measurements as they are used to support carbon trading and offsetting is that it is very hard, if not impossible, to know how long the carbon in the soil is going to stay there. The carbon sequestered needs to stay out of the atmosphere for long enough to have an effect on global climate change. Therefore, soil carbon measurements need to happen across a suitable duration of time.

Again, there are sub-problems here. First, it is uncertain how long the carbon would *need* to stay in the soil to have the intended effect; the goal is unclear. Is 50 years enough? 100?⁹ Climate change is being caused by removing carbon from the lithosphere (via burning fossil fuels) and putting it into the atmosphere, and some argue that putting it back in the biosphere (e.g., via photosynthesis) is a short-term, band-aid solution [115]. Soils are situated between the lithosphere and the biosphere, containing both mineral and organic matter. Soil carbon, likewise, can be organic or inorganic. All this is to say that there are some questions that remain to be answered about what kind of carbon storage “counts” for carbon to be considered sequestered by soils.

Second, it is uncertain what would or would not cause sequestered carbon to stay in the soil for long enough, once it is there. For example, say a farmer is paid via a carbon offsetting program to do a no-till treatment, and she abides by her contract and does not till a certain area of her land. How long does she have to not till it for, to retain the carbon in the soil? Does she have to maintain regenerative agricultural practices for the entire duration of the promised sequestration period, or can she stop after she is no longer being paid to use these practices?

Third, and more pertinent for measurement, there is uncertainty about just how long, in fact, carbon in the soil stays there, once it is there. Of course, one can take many measurements over time and can thereby see soil carbon concentrations fluctuating to various degrees. But because there are many possible causes of these fluctuations besides just the management of the soil, it is difficult to know whether the very same carbon that was sequestered due to some intentional management practice – even if we did know that quantity – is still there after any given amount of time, or whether it is no longer there and has, perhaps, been replaced with other carbon (of atmospheric origins or otherwise).

⁸ Thanks to Mauricio De La Parra Gurr for calling this to my attention.

⁹ Apparently, Indigo promises 100 years, but this number is not realistic [92].

Again, for a carbon trading or offsetting program, we should only “count” carbon as sequestered if it has been removed from the atmosphere by the intentional management practice, and would not have been removed otherwise. The difficulties assessing the counterfactual claim here also, then, apply to assessing that claim over time.

In summary, supporting carbon trading and offsetting schemes using measurements of soil in carbon requires that we know how long the carbon stays in the soil, relative to alternative amounts of time and alternative reasons why more or less carbon might be in the soil. There are various uncertainties associated with these claims.

These three sources of uncertainty – related to selecting the appropriate sampling depth, selecting and assessing counterfactuals, and measuring the longevity of carbon sequestration – are just a taste; other sources of uncertainty might include geographic variation (e.g., see [87, 92,116]) or other issues related to sampling (e.g., how frequently in space and time to sample a particular land area). And we have not even dealt at all with the sources of uncertainty within the various measurement instruments used (e.g., see [117] for discussion of the limitations of the standard, dry combustion method and [118] for discussion of a recently-developed instrument that can be used in the field for this purpose). Given what the measurements of carbon in soil are for – assessing the efficacy of carbon trading and offsetting schemes at incentivizing soil management practices that sequester carbon – all of these sources of uncertainty combine together in a complex process of measurement as aim-directed inquiry, as a pragmatist view of measurement would suggest [1,3,42,44,102,119].

Suffice it to say, measurements that estimate carbon in soil are highly uncertain. Before turning to an assessment of how that uncertainty affects these measurements’ ability to be used for particular purposes, I would first like to acknowledge some of the work that soil scientists are doing to manage these sources of uncertainty.

3.2. Strategies for mitigating uncertainty

Of course, it is not as though these sources of uncertainty are going unaddressed: many soil scientists are savvy to these sources of uncertainty and have been working on creative ways of mitigating them in order to produce more reliable measurements of soil carbon.

The main strategy for mitigating the sources of uncertainty detailed above involves supplementing the use of measurements of carbon with the use of models that help predict or adjust the carbon data. Using models to correct data is a common strategy, especially in scientific contexts where uncertainty is pervasive [57,120]. For example, in climate science, climate simulations are used not only to project the future state of the climate under different emissions scenarios, but also to process existing data about the climate in order to be able to use those data to reconstruct and understand past climate trends and make causal attributions for those trends [55,56]. In general, using models to help mitigate the problems with data and measurement is a common practice in science, especially when there are impediments to dealing with those problems more directly.

The soil carbon case is no exception. Models are often used to refine estimates of how much carbon is in a certain amount of soil, and to forecast those estimates into the future, based on a theoretical understanding of the forces that would cause carbon (and other elements like nitrogen) levels in soil to fluctuate (e.g., [121,122]). Some of these models are expressly used to help with the kinds of carbon accounting needed for sequestration programs [123].

However, although the use of models to correct for the deficiencies of data is a common and effective practice, introducing models into the mix does not come without its own challenges. For example, “leading soil models are calibrated to results of soil studies in temperate countries such as the U.S. and Europe, where most studies have historically been done” [93], part of a larger problem in many areas

of ecology [124].¹⁰ Furthermore, there are some who worry that the models’ inability to explicitly represent some of the more complex behaviors of soils means that these models run the risk of producing inaccurate estimates and predictions [92,125].

Of course, using an adequacy-for-purpose framework for model evaluation [30], we should not go so far as to say that these idealizations or abstractions incorporated into the models make them bad – idealizations and abstractions are often exactly what make models useful, and producing a more representationally accurate model sometimes means producing a model that is less fit for its intended purposes [35–37]. In other words, models of soil need not necessarily represent soil in its full complexity to be useful (cf. [93]). However, in this case, there is reason to believe that at least some of the models are not helping the situation with measurement uncertainty much, with estimates of model uncertainty sometimes so high that the model is not able to differentiate increases from decreases in carbon [126].¹¹ For other models, it is hard to even quantitatively estimate the uncertainty at all, although there is even some reason to believe that the models systematically overestimate the amount of carbon stored in the soil [92,93]. Including quantitative uncertainty estimates in versions of these models used by farmers and other stakeholders has been a major project of the last few years, and only this year (2025) have uncertainty estimates and an explanation of the methodology used to calculate them been added to COMET-Farm (the main greenhouse gas inventorying model used by farms across the United States) [127].

3.3. Purposes of soil carbon measurement

We can now at least say that measurements of carbon in soil are highly uncertain, and current efforts to mitigate that uncertainty, including the use of models to supplement these measurements, have not been fully successful at doing so. However, just saying that a measurement is uncertain does not immediately tell us that it is inadequate to serve a particular purpose, although uncertainty estimates are “essential for judging the adequacy of data” [43]. Measurements (like models, data, and other scientific tools) can have or lack any number of features that facilitate or hinder their ability to serve a purpose, and uncertainty is just one of those features. Whether uncertainty is a hindrance has to be evaluated in a contextual manner.¹²

So, are these measurements of soil carbon adequate or inadequate for a purpose? Well, it will depend on the purpose we specify. For instance, if the measurements are intended to help facilitate actual carbon sequestration, the presence of so much uncertainty makes it unclear whether this purpose is being served adequately. If the measurements are intended to help support programs that incentivize certain interventions, such as regenerative agricultural practices, regardless of the

¹⁰ Thanks to Alisa Bokulich for calling this issue to my attention.

¹¹ Just to emphasize this a bit: in case one thought that it was enough to just know that some soil treatments sequester carbon, without knowing *how much* carbon they sequester, the magnitudes of the uncertainty here show that sometimes we do not even know whether *any* carbon is being sequestered, or even if some carbon is being emitted. Thanks again to Roberta Millstein for encouraging me to emphasize this point.

¹² As an anonymous referee helpfully pointed out, it would be nice to be able to specify precisely what effect the current levels of uncertainty have on each of these purposes and what effect reducing that uncertainty would have. However, there are several barriers to performing this analysis quite so precisely. First, as mentioned above, many of uncertainties discussed in Section 3 are not quantified as of yet. Second, even if they were, tracing the effect of that uncertainty through to its relationship to a more-ultimate or even a more-proximate goal would be quite the undertaking, although a worthwhile one. That said, in this paper we only need to see how this case illustrates a broader point about the adequacy-for-purpose view of measurement evaluation, which does not require making quite so precise of connections between degrees of uncertainty and their effects.

eventual effect of these practices, they are perhaps more adequate for doing so. And if the measurements are intended to facilitate carbon trading and offsetting programs and benefit the corporations that run or participate in these programs, then, again, these measurements seem perfectly adequate for doing so, despite their associated uncertainties.¹³ In particular, the uncertainty involved in measuring carbon in soils is not limiting these corporations' ability to *claim* carbon offsets at all.

For two of these candidate purposes, then – incentivizing certain soil management practices or facilitating carbon trading and offsetting programs, whether or not either of these result in actual sequestered carbon – an adequacy-for-purpose framework for measurement evaluation, as currently conceived, is not going to provide us with any tools to say that something has gone wrong. The highly uncertain measurements are perfectly adequate for serving those purposes. However, some may find this verdict unsatisfactory: certainly, there would be something criticizable here! In the next section, I will argue that levying that criticism requires further analysis of the purposes for which carbon trading and offsetting are being pursued.

4. Adequacy for more-ultimate purposes

Let us take stock. Measurements of carbon in soil are highly uncertain, for several reasons. Some of these sources of uncertainty might raise suspicions that the carbon trading and offsetting schemes which rely on these measurements to sell carbon credits are not doing what they set out to do. Yet, according to the adequacy-for-purpose method for evaluating measurements, perhaps nothing has gone wrong – there are at least some purposes, such as incentivizing regenerative agricultural practices or propping up carbon trading and offsetting programs, that can be achieved using these measurements, regardless of the uncertainties involved.

However, as I will demonstrate in this section, if we look beyond the adequacy-for-purpose view and evaluate these measurements in a more thoroughly pragmatist fashion, it is still possible to point to what has gone wrong with these measurements. In order to do so, we first have to distinguish between more-proximate and more-ultimate goals, and then we have to specify and assess the more-ultimate goal behind the carbon trading and offsetting programs. The difference between more-proximate and more-ultimate purposes can be thought of as follows: “more-proximate purposes are often pursued in service of more-ultimate purposes, and these more-ultimate purposes might have practical upshots that constrain which more-proximate purposes should be pursued” [52, p. 19].¹⁴ In other words, purposes in science often have a layered structure such that the more-proximate purpose of using a scientific tool (model, measurement, etc.) is pursued only insofar as it, in turn, serves a more-ultimate purpose. My contention is that we have to detect and evaluate the more-ultimate aims of soil carbon measurement in order to pragmatically assess these measurements.

Once we specify the more-ultimate purposes of these measurements, we then find ourselves in one of two scenarios, and can levy a different kind of criticism in each scenario. First, if the more-ultimate goal is nefarious, we can criticize it for being that. Second, if the more-ultimate goal is laudable, we can assess whether the measurements are serving that laudable, more-ultimate goal (even if they are serving some more-proximate goal).

¹³ One might even go so far as to say that the uncertainties themselves are facilitating these programs, a stronger proposition than just saying that the uncertainties are not hindering these programs. I am sympathetic to this idea but substantiating it would go beyond the scope of this paper.

¹⁴ The main contention of [52] is that the tools being used to pursue the more-proximate purposes – in the case at hand, the data scientists have at their disposal – can also constrain the more-proximate purposes. This may also be going on in the soil carbon case, as well, but I leave that to the side for the purposes of this paper.

Of course, one of the goals of criticizing these practices would be to reform them, a project beyond the scope of this paper but a crucial one nonetheless. For example, the incentive structures encouraging carbon crediting as we know it might need to be adjusted, an endeavor that would require the input of economists (e.g., see [128,129] for some relevant ideas).¹⁵

4.1. Nefarious more-ultimate goal: Financial gain

Corporations that can claim to have offset their carbon emissions can appeal to customers that demand they do so, and middle-man companies that facilitate carbon offsetting get to make money off of doing so. Although some corporations may have a mixed motivation to both do some environmental good and also make money, it would be unsurprising if some of them are purely motivated by financial gain.¹⁶ A purely financial motivation for participating in carbon trading and offsetting schemes would involve considering the schemes successful if they facilitate profiting financially, without regard for whether these schemes also result in environmental benefit (or harm).

If the more-proximate goal of supporting carbon trading and offsetting programs is only being pursued for financial gain, regardless of any environmental consequences, this more-ultimate purpose is itself criticizable as nefarious. Indeed, we might even be able to say that the *reason* that the uncertainty of the measurements of soil carbon do not impede pursuit of the more-proximate goal is that the more-proximate goal is only pursued as a means to a nefarious more-ultimate goal. Additional details of the criticism of the more-ultimate goal would have to be spelled out, but it is not difficult to imagine these details getting filled in. Focusing on a nefarious more-ultimate goal allows us to criticize the research practices here, without relying on an adequacy-for-purpose check to do so.

4.2. Laudable more-ultimate goal: Sustainability

There is an obvious alternative more-ultimate goal for setting up carbon trading and offsetting programs: the sustainability-related goal of mitigating climate change via carbon sequestration. I will take it for granted that this goal is at least less criticizable than one related to financial gain (some general worries about carbon sequestration notwithstanding).

However, with sustainability in place as the more-ultimate goal, we can see that even though the uncertain measurements of soil carbon may be perfectly adequate for pursuing the more-proximate goal of supporting carbon trading and offsetting programs, these uncertain measurements are *not* adequate for pursuing the more-ultimate goal of actually sequestering carbon. In particular, if the carbon trading and offsetting schemes are able to claim that more carbon is being sequestered than is actually being sequestered, as some are worried that they can, then although the carbon trading and offsetting schemes are functioning fine as evidenced by their continued existence and popularity, they are not functioning to actually serve the more-ultimate goal of climate change mitigation. And it is because of the vast amounts of uncertainty related to the soil carbon measurements that they fail to achieve that goal, making these measurements inadequate for the more-ultimate purpose even if they are adequate for the more-proximate one.

¹⁵ Thanks to an anonymous referee for suggesting this as a future direction and the connection to this literature.

¹⁶ Discussions of “greenwashing” often criticize corporations for just this [130].

5. Conclusion

The foregoing analysis leaves three options open for the proponent of an adequacy-for-purpose method for measurement evaluation, each of which might be applicable to further cases than just to those related to soil carbon.

First, they can simply say of this case, “yes, the measurements are adequate for their purposes and nothing has gone wrong.” This is what some of the existing literature on adequacy-for-purpose might suggest. Even if we add an analysis of more-ultimate purposes into the mix, on this view we can still say, “the measurements are adequate for the more-proximate purposes, and the more-proximate purposes serve the more-ultimate purposes, so nothing has gone wrong”.

I think that this view is unsatisfactory, because it does not enable us to levy criticism even if we were sure that the more-ultimate purposes being served were nefarious. If we want to maintain our ability to criticize scientific practices, as we often do [131], then we need to have more avenues at our disposal. I have proposed two remaining routes that go beyond what the first has to offer. The second is to identify more-ultimate purposes as nefarious, and criticize them as such on independent grounds. No adequacy-for-purpose evaluation is needed here, although more work would need to be done to figure out exactly what the “independent grounds” for criticism might be.

Third, though, we can even grant that practitioners in this case have laudable, more-ultimate aims, and utilize an adequacy-for-purpose assessment to say that the measurement practices are inadequate for the more-ultimate purposes, even if they are adequate for the more-proximate ones. Expanding the adequacy-for-purpose account in this way requires that we think more broadly about the range of purposes for which tools like measurement are developed and deployed, beyond just their proximate purposes, a project consonant with broader insights in a pragmatist philosophy of measurement.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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